DV INTERVIEW · GLANCE SHEET 3

AXI Write & Read Transfers, Cycle by Cycle

How a burst actually moves on the wires: handshakes, stalls, last beats, responses and byte lanes.

1 signal HIGH
⇄ handshake (transfer happens)
0 signal LOW / idle

W Write burst — 4 beats, INCR, 32-bit bus, AWADDR = 0x00

`AWLEN = 3` · `AWSIZE = 3'b010` (4 B) · `AWBURST = 2'b01` · slave stalls the write data once (WREADY low in T4).

ACLK	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
AWVALID	1	⇄	0	0	0	0	0	0	0	0
AWREADY	0	⇄	0	0	0	0	0	0	0	0
AWADDR	0x00	0x00	–	–	–	–	–	–	–	–
WVALID	0	0	⇄	1	⇄	⇄	⇄	0	0	0
WREADY	0	0	⇄	0	⇄	⇄	⇄	0	0	0
WDATA	–	–	D0	D1	D1	D2	D3	–	–	–
WLAST	0	0	0	0	0	0	1	0	0	0
BVALID	0	0	0	0	0	0	0	1	⇄	0
BREADY	0	0	0	0	0	0	0	0	⇄	0
BRESP	–	–	–	–	–	–	–	OKAY	OKAY	–

T2 — address accepted (AWVALID & AWREADY).

T3, T5, T6, T7 — 4 data beats. **T4** = slave back-pressure: master **holds D1** and WVALID until accepted.

T7 — WLAST with beat 4 (= AWLEN + 1).

T8–T9 — slave raises BVALID after WLAST; master accepts at T9 with BREADY → write done.

R Read burst — 4 beats, INCR, 32-bit bus, ARADDR = 0x40

`ARLEN = 3` · `ARSIZE = 3'b010` · `ARBURST = 2'b01` · the **master** stalls once (RREADY low in T6).

ACLK	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
ARVALID	⇄	0	0	0	0	0	0	0	0	0
ARREADY	⇄	0	0	0	0	0	0	0	0	0
ARADDR	0x40	–	–	–	–	–	–	–	–	–
RVALID	0	0	0	⇄	⇄	1	⇄	⇄	0	0
RREADY	1	1	1	⇄	⇄	0	⇄	⇄	1	1
RDATA	–	–	–	D0	D1	D2	D2	D3	–	–
RRESP	–	–	–	OK	OK	OK	OK	OK	–	–
RLAST	0	0	0	0	0	0	0	1	0	0

T1 — address accepted in the same cycle (VALID & READY together).

T2–T3 — slave latency; RREADY can sit HIGH waiting (allowed).

T6 — master back-pressure: slave **holds D2** and RVALID until RREADY returns.

T8 — RLAST on beat 4 (addr 0x4C). No separate response channel — RRESP came with every beat.

1 Single-beat transfer (AWLEN = 0)

- One AW handshake, one W beat with **WLAST = 1**, one B response.
- AW and W can handshake in the **same cycle**, or W can even come **first**.
- Fastest write = 2 cycles (AW+W together, then B). Fastest read = AR, then R next cycle.
- AXI-Lite is always this case — no LEN, no LAST, no ID.

WRITE ORDERINGS THE SLAVE MUST HANDLE

AW then W then B — normal

AW + W same cycle then B — fastest

W then AW then B — legal, often missed in TBs

2 Overlapping (outstanding) transfers

Master issues the next address before the previous burst's data finishes — this hides slave latency.

Cycle	T1	T2	T3	T4	T5	T6	T7	T8
AR	A	B	–	–	–	–	–	–
R	–	–	A0	A1	A2	A3·L	B0	B1·L

- Without overlap B's address would wait until T7 → 4 wasted cycles.
- Same ID → B's data must follow A's. Different IDs → B's data could return first.
- Max outstanding is a design parameter ("acceptance capability") — cover 0 … max in DV.

3 Byte lanes: narrow & unaligned transfers (32-bit bus = 4 lanes)

NARROW INCR · SIZE = 1 B · START 0X00 · 4 BEATS

	B3	B2	B1	B0	
beat 1				B0	0001
beat 2			B1		0010
beat 3		B2			0100
beat 4	B3				1000

Narrow = SIZE smaller than bus. Active lane **moves** each beat with the address.

UNALIGNED INCR · SIZE = 4 B · START 0X01 · 3 BEATS

	B3	B2	B1	B0	
0x01	B3	B2	B1		1110
0x04	B3	B2	B1	B0	1111
0x08	B3	B2	B1	B0	1111

Only the **first beat** is unaligned; later beats snap to aligned addresses (0x04, 0x08). Total = 11 bytes.

RULES TO REMEMBER

- Lane for a byte = `addr mod (bus bytes)` (little-endian).
- INCR next address = `aligned(addr) + 2^SIZE`.
- **FIXED** narrow: the same lanes every beat.
- **WRAP** must start aligned — unaligned WRAP is illegal.
- Reads use the same lane logic; master just picks the bytes it needs.
- WSTRB must be 0 on lanes outside the transfer.

4 What to check & cover for transfers

CHECKS (MONITOR / SVA)

- Payload (WDATA, WSTRB, RDATA…) **stable while VALID && !READY** — the T4 / T6 stall cases above.
- Exactly LEN+1 beats; LAST only on the final beat.
- BVALID never before WLAST handshake; RVALID never before AR handshake.
- WSTRB lanes match address/SIZE for narrow & unaligned beats.
- Read data matches reference memory; write updates only strobed bytes.

COVERAGE

- Back-pressure on every channel (READY low for 0, 1, N cycles).
- W before AW, same-cycle AW+W, AW before W.
- Back-to-back bursts (no idle), overlapped addresses, max outstanding.
- Narrow × aligned/unaligned × burst type × LEN.
- Error responses mid-burst on R (RRESP changes beat to beat).

? Transfer interview Q&A

Question	Crisp answer
Minimum cycles for a single write?	2 — AW+W handshake together, B next cycle.
What happens when WREADY drops mid-burst?	Master holds the same WDATA/WSTRB and WVALID until WREADY returns.
Can RREADY be held HIGH all the time?	Yes, and it's common; it just means "always ready".
How does the slave know the burst ended?	WLAST (it may also count LEN+1).
Can a read error stop the burst early?	No — all LEN+1 beats are still returned with RRESP error.
Start 0x03, SIZE 4 B, 2 beats — strobes?	Beat 1: 1000, beat 2 (0x04): 1111.
Why overlap addresses?	Hides latency; keeps the data channel busy back-to-back.
Can AW for burst 2 come before W of burst 1 ends?	Yes — outstanding writes; in AXI4 W data still follows AW order.
Who drives BREADY / RREADY?	The master.